American Computer Science League

2025-2026 ● Contest 4: Short Problems Solutions ● Intermediate Division

1. Graph Theory

The adjacency matrix raised to the 2nd power for paths of length 2 to get a 3-scoop cone is as follows:

	WB	TR	RB	CB	BA
WB	0	1	1	1	1
TR	0	0	1	1	1
RB	0	0	0	1	0
CB	0	0	1	0	0
BA	0	0	0	1	0

2 =

	WB	TR	RB	CB	BA
WB	0	0	2	3	1
TR	0	0	1	2	0
RB	0	0	1	0	0
CB	0	0	0	1	0
BA	0	0	1	0	0

There are 12 ways to do it:
WB TR RB, WB TR BA (C), WB TR CB, WB CB RB, WB BA CB (D), WB RB CB, TR RB CB, TR BA CB, TR CB RB (A), RB CB RB, CB RB CB, and BA CB RB (B). RB, TR, BA (E) isn't possible.

E. Roasted Beet, Tart Rhubarb, Baked Alaska

2. Graph Theory

The only cycles that exist are of length 2: ABA, CHC. The vertices that are not used are D, E, F, G, which is 4 of them.

B. 4

3. Digital Electronics

The equivalent Boolean algebra expression is: $\overline{AB} \oplus ((\overline{B + C})C)$.
It simplifies as follows:

$\overline{AB} \oplus ((\overline{B + C})C)$
$= \overline{AB} * \overline{(\overline{B + C})C} + \overline{\overline{AB}} * (\overline{B + C})C$
$= \overline{AB} * (B + C + \overline{C}) + AB * \overline{B} * \overline{C} * C$
$= \overline{AB} * 1 + 0$
$= \overline{AB}$

That means there are 6 of them, not (1,1,*).

D. 6

4. Digital Electronics

The Boolean algebra expression represented by the following digital circuit is: $\overline{\overline{A}} * \overline{(\overline{A + B + BC})}$

$= \overline{\overline{A}} + \overline{\overline{A + B + BC}}$

$= A + A + B + BC$

$= A + B(1 + C)$

$= A + B * 1$

$= A + B$

D. $A + B$

5. Assembly Language

```
A   DC      7
B   DC      8
    LOAD    B
    SUB     =10
    MULT    =-1
    STORE   C
    LOAD    B
    MULT    C
    STORE   D
    LOAD    A
    MULT    B
    MULT    =100
    ADD     D
    STORE   ANS
    PRINT   ANS
    END
```

This program makes A = 7, B = 8, and C = 10-B = 2. Then D = B*C = 16 so ANS = A*B*100+D = 7*8*100+16 = 5600+16 = 5616. This is also the value of 78 * 72 and the process works for any digits A, B, and C.

E. 5616

6. Assembly Language

```
ANS     DC      0
        READ    P
TOP     READ    N
        LOAD    N
        BE      STOP
        LOAD    P
        SUB     N
        BL      CONT
        LOAD    ANS
        ADD     =1
        STORE   ANS
CONT    LOAD    N
        STORE   P
        BU      TOP
STOP    PRINT   ANS
        END     ANS
```

This program counts how many "step-downs" there are, in other words, how many times the number is less than or equal to the previous value. These occur when the input is 11, 13, and 7.

D. 3